ServiceNow Certified Application Developer

SmartInternz

Project Title

Automated Network Request Management in ServiceNow

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Project overview

The **Automated Network Request Management in ServiceNow project** focuses on simplifying and accelerating the handling of network-related service requests. By leveraging ServiceNow's service catalog, workflows, and automation capabilities, the solution reduces manual effort and ensures that requests are processed efficiently. Users can submit network-related requests through a self-service interface, while automated routing, approvals, task assignments, and notifications streamline the fulfillment process. The system improves request tracking, reduces processing delays, and provides better visibility for both users and IT teams.

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1. Introduction

Modern organizations rely heavily on efficient network services to support their day-to-day operations. As businesses grow, the number of network-related service requests, such as access requests, configuration changes, connectivity issues, and infrastructure support, also increases. Managing these requests manually can lead to delays, errors, repeated follow-ups, and limited visibility into the request lifecycle.

To address these challenges, the Automated Network Request Management in ServiceNow project provides a streamlined solution for managing network-related service requests. The system uses ServiceNow's service catalog, workflow

automation, approval processes, task assignments, and notifications to simplify request handling.

Users can submit requests through a self-service interface, while the system automatically routes requests to the appropriate teams, manages approvals, assigns tasks, and provides status updates. This reduces manual effort, improves processing accuracy, and provides better transparency for both requesters and IT teams.

The project demonstrates how ServiceNow can be used to automate and manage IT service processes in a structured and efficient manner.

2. Project Objectives

The main objective of the Automated Network Request Management in ServiceNow project is to develop a structured and automated system for managing network-related service requests. The project aims to reduce manual effort, improve request processing, and provide better visibility throughout the request lifecycle.

The system allows users to submit network service requests through a self-service interface. Based on the request details, the system automatically routes the request to the appropriate team, initiates approval processes, assigns tasks, and sends notifications to the concerned users.

The key objectives of the project are:

• To automate the submission and processing of network service requests.

• To reduce manual effort and minimize errors in request handling.

• To implement automated approval and task assignment processes.

• To provide notifications and status updates to users.

• To improve transparency and tracking of requests.

• To demonstrate the use of ServiceNow workflows, service catalog, and automation features.

3. Key Features

- Custom service catalog for network-related requests

- User-friendly forms for capturing request details
- Automated approval workflow for network requests
- Automatic task assignment to the appropriate team
- Email notifications for request updates and approvals
- Real-time tracking of request status
- ServiceNow Flow Designer for process automation
- Improved transparency and reduced manual effort

4. ServiceNow Developer Setup:

Access the ServiceNow Developer Portal

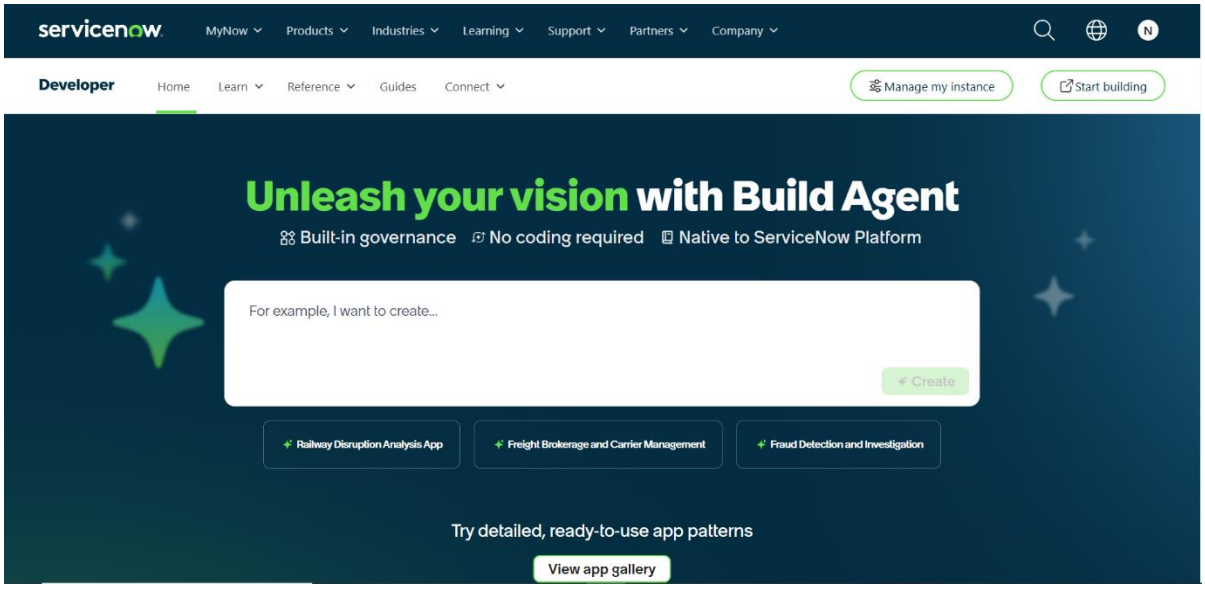
1. Go to ServiceNow Developer Portal(https://developer.servicenow.com/dev.do).

Sign up for a free developer account and fill out the following details.

2. After signing up, you will get a verification mail to your provided email id. After the verification your ServiceNow Developer Portal Home Page will appear.

3. Now click on start building it will take you to the section where you can request a Personal Developer Instance (PDI) or start using App Engine Studio and other tools.

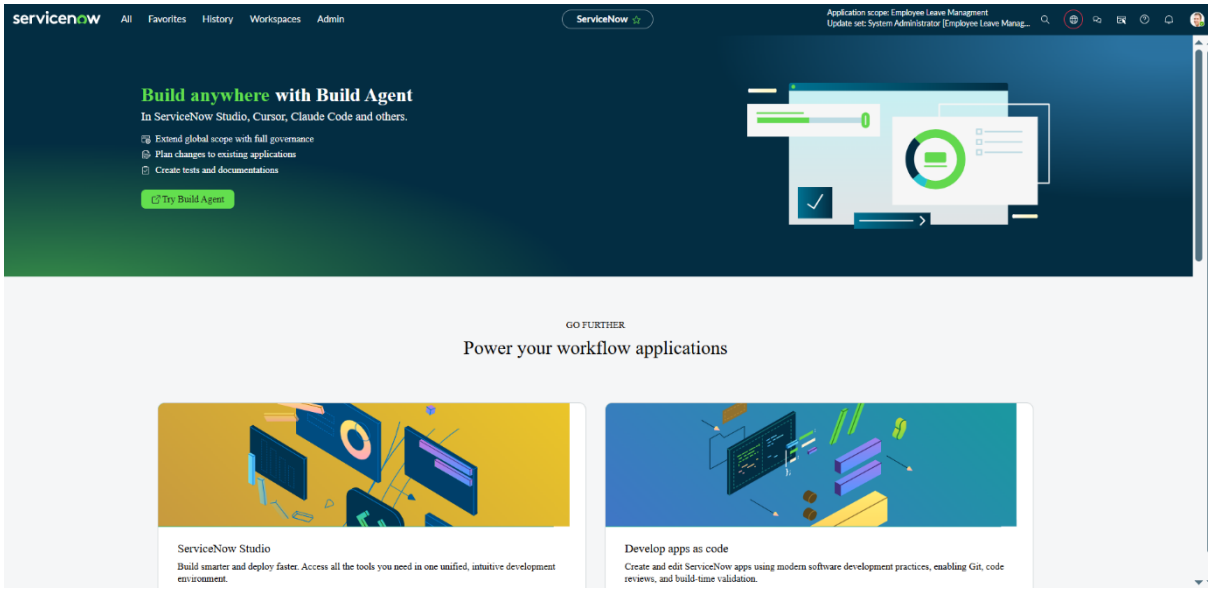
4. **Profile Icon(Top Right Corner)** → Manage your account, request instances, and check your developer profile.



5. Project Implementation in ServiceNow:

After the instance building is completed, the page will be directed into your Creator Studio.

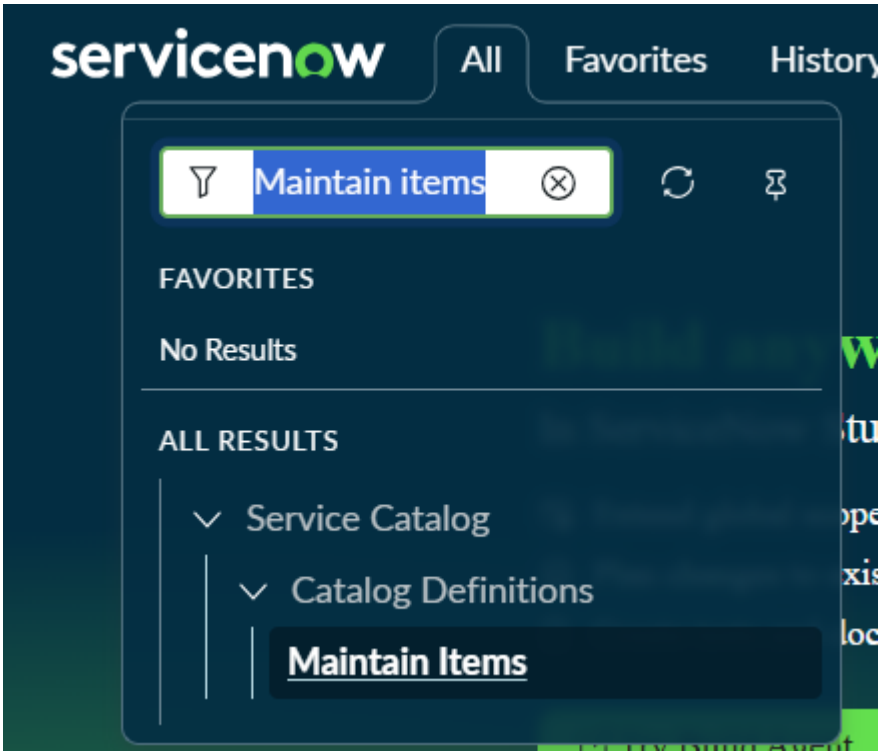
Creator Studio in ServiceNow provides a **guided, no-code environment** to build applications quickly. It is especially used for creating **request-based applications** by defining forms, setting up tables, and automating workflows.



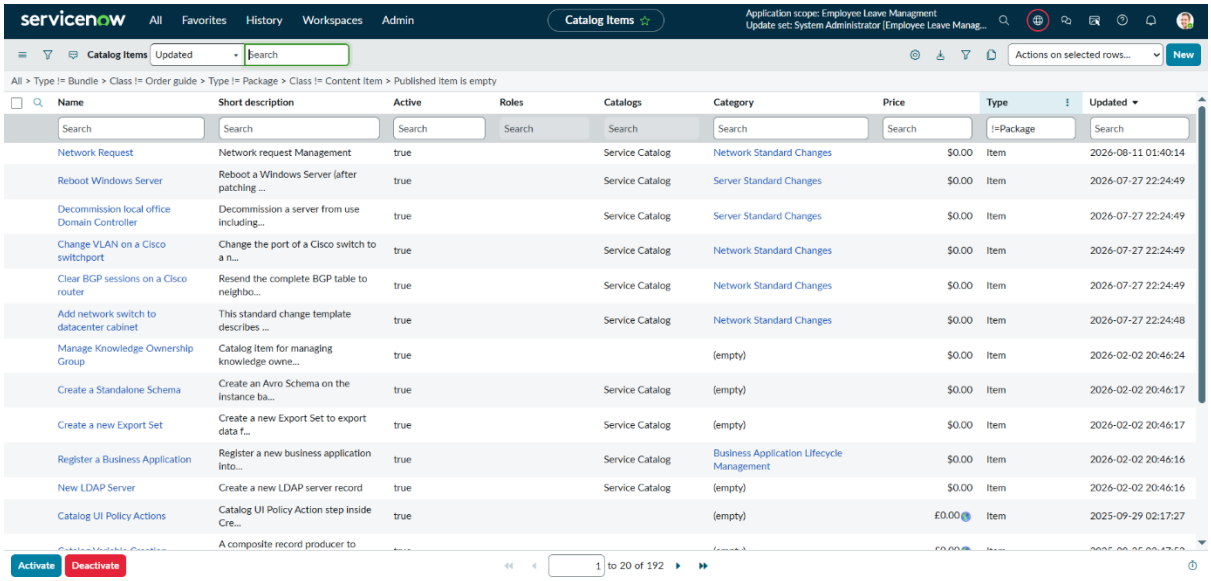
a. Service Catalog Creation

i: Catalog Creation

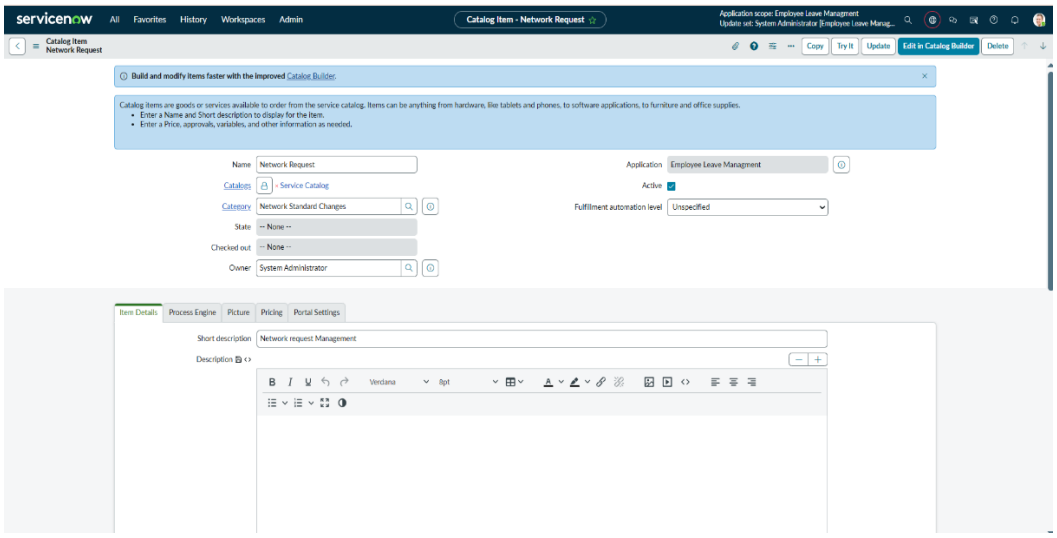
1. Navigate to Application navigator
2. Click on All >> search for Service Catalog
3. Under Service Catalog >> Maintain items
4. Click on New
5. Fill the details >> Name – Network Request
6. Select Catalog >> Service Catalog
7. Select Category >> Network Standard Changes
8. Fill the Short Description as Network Request Management.
9. Click on Save.



Click on New



Create the Network Request



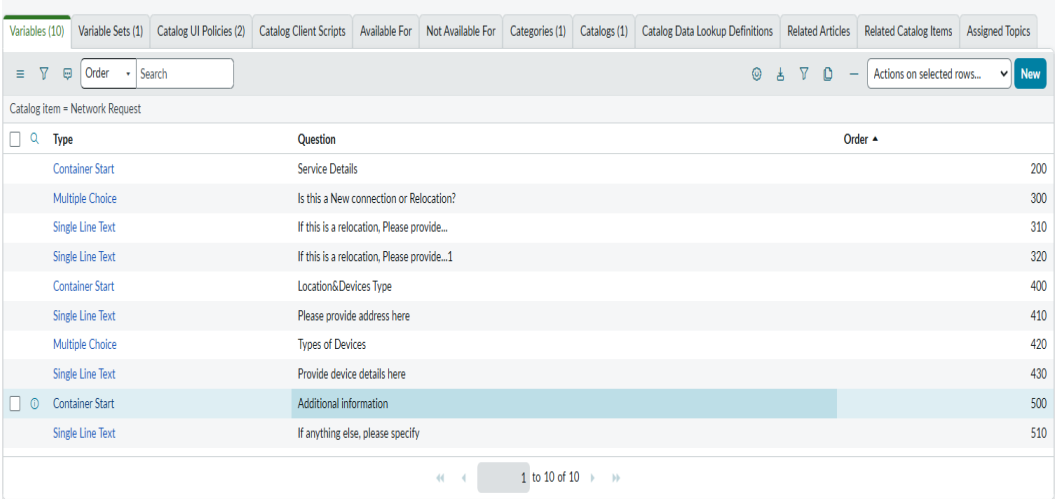
ii: Variables Configuration

Open the catalog item just created.

Scroll down to the **Variables** related list and click **New** to create form fields.

1. Select Variables type as Single line text, Multi line text, Reference, Container Start etc as per requirement
2. Catalog Item – Network Request
3. Order – 100,200,300,,,
4. Question – Provide the variable label

5. Name – Provide the variables name(used for scripting)
6. Tooltip – This will appear when cursor overed on the field
7. Example text – This will suggest what we need to enter on the field.
8. Mandatory, Read-Only – Need to configure on demand.
9. Auto populate – Need to select dependent variable, apply dot walking to get selected value.
10. Click on Save or submit.



iii: Variables Types

1. Is this a New connection or Relocation? >> Choice >> New/ Relocation/None

2. If this is a relocation, Please provide your relocated address here>>String

3. Types of devices>> Choice>> Laptop/Mobiles/Others

4. Please provide address here>>String

5. Provide device details here>> String

6. If anything else, please specify>> String

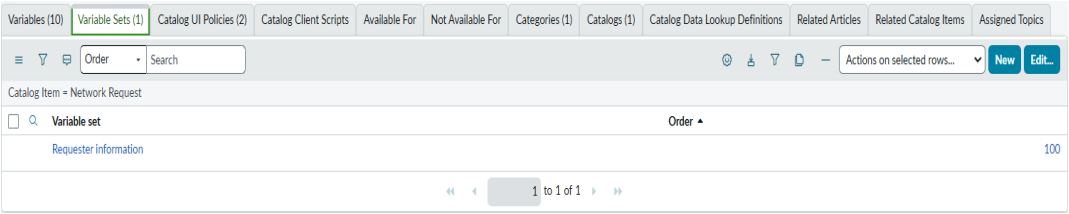
iv: Variable Set Configuration

To enhance form usability:

■ Navigate to the Variable Sets (optional).

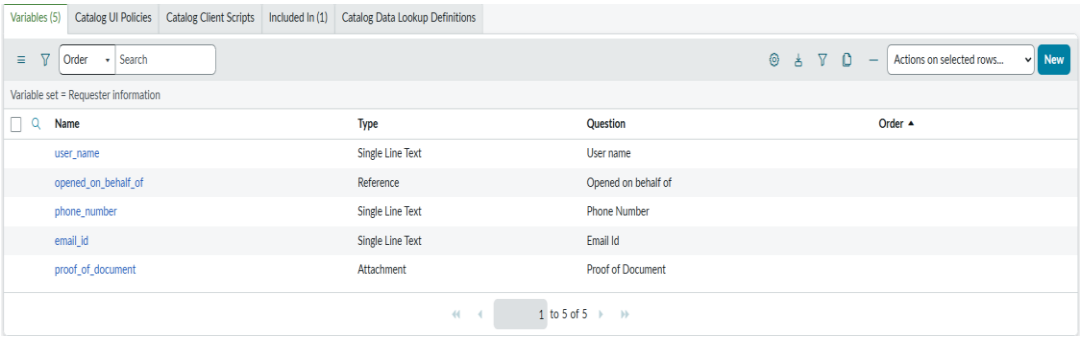
■ Follow the same procedure as we used for Variables Creation, for the variable set as well.

■ Apply variable sets to the catalog item.



Variables Types

1. Opened on behalf of >> Reference>> reference to user table
2. Email Id >> Single line text >> Auto populate by Opened on behalf of variable.
3. User name >>Single line text >> Auto populate by Opened on behalf of variable.
4. Phone Number >>Single line text >> Auto populate by Opened on behalf of variable.
5. Proof of Document >> Attachment



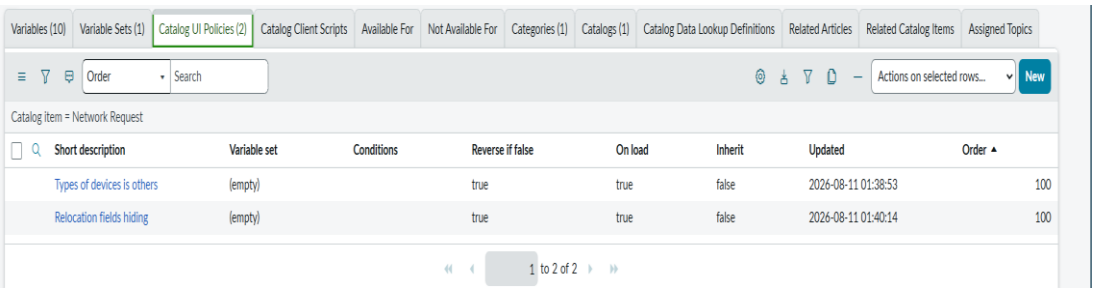
V: Catalog UI Policy Configuration

Scenario: If user selects types of devices is **Others**, then Please specify field should populate.

Procedure:

1. Navigate to catalog items
2. Open Network Request item

3. In related list, we have Catalog UI policy
4. Click on New button to configure New UI policy
5. Select Applies to as Catalog item
6. Select catalog item as Network Request
7. Provide short description, if required
8. Apply condition>> **types of devices** is **others**
9. Click on save, after saving the form will get UI policy actions in the related list
10. Click on New button to configure new UI Policy action, and Select the variable which we
want to display on condition
11. Make Visible True as per our requirement
12. Update the UI Policy and Test the same on Catalog form



b. Creation of Table

i. Creation of Table

1.Navigate to: System Definition > Tables.

a.Click New to create a new table.

b.Fill in Table Information:

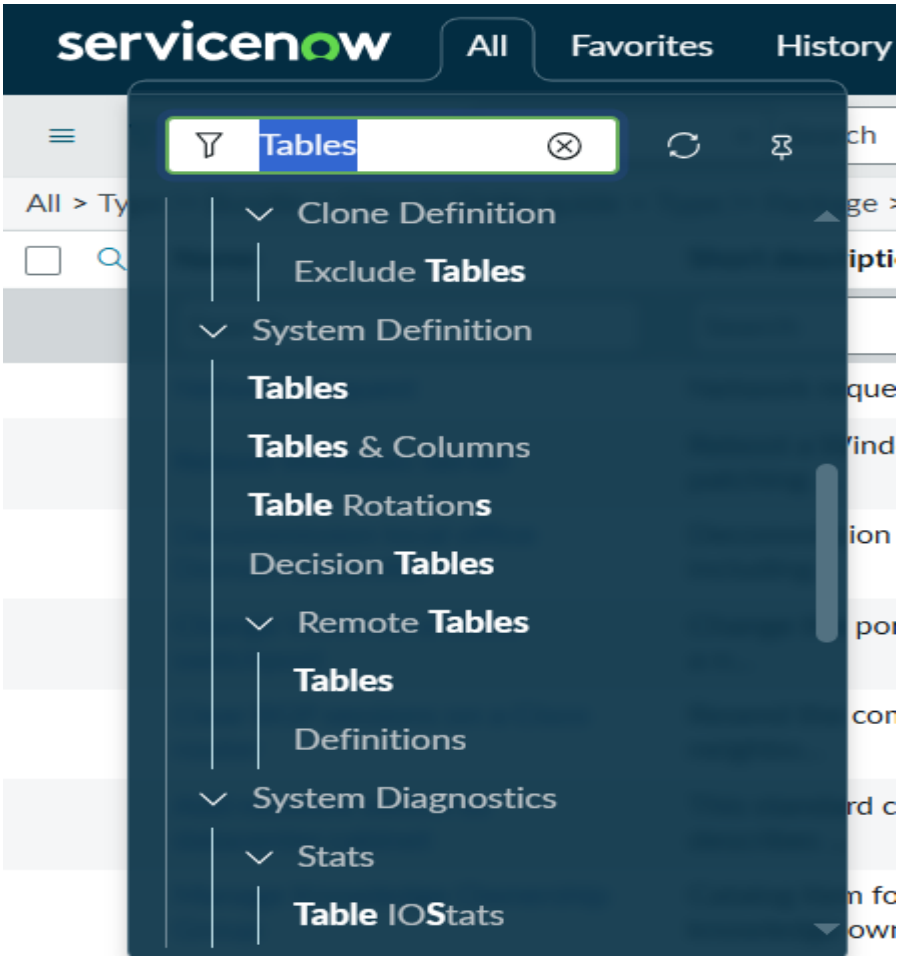
i.Name: Name of the table ------

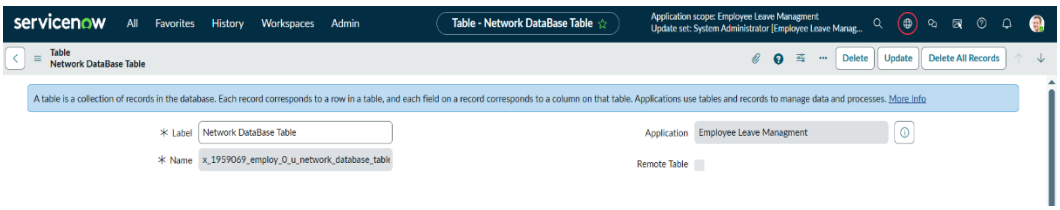
ii.Label: Backend name of the table------

iii.Auto-generate schema: Leave it checked if you'd like ServiceNow to

auto-generate schema fields.

c.Click Submit to create the table.





ii. Creation of fields

In ServiceNow, fields are created at the **table** level. To create a field, you first need to identify the table where the field will reside.

1. In the **Application Navigator** (left-side panel), type **Tables** in the search bar.
2. Under **System Definition**, click **Tables**. This will take you to a list of all tables in the system.

a. Select the Table to Add the Field

From the list of tables, search for and select the **table** you want to add a field to. For example, if you want to add a field to the **Network database** table:

1. Type "**Network database**" in the search box or scroll through the list.

2. Click on **the Network database** table name. You'll now see a list of all fields (columns) associated with the **Network database** table.

b. Open the Table's Columns

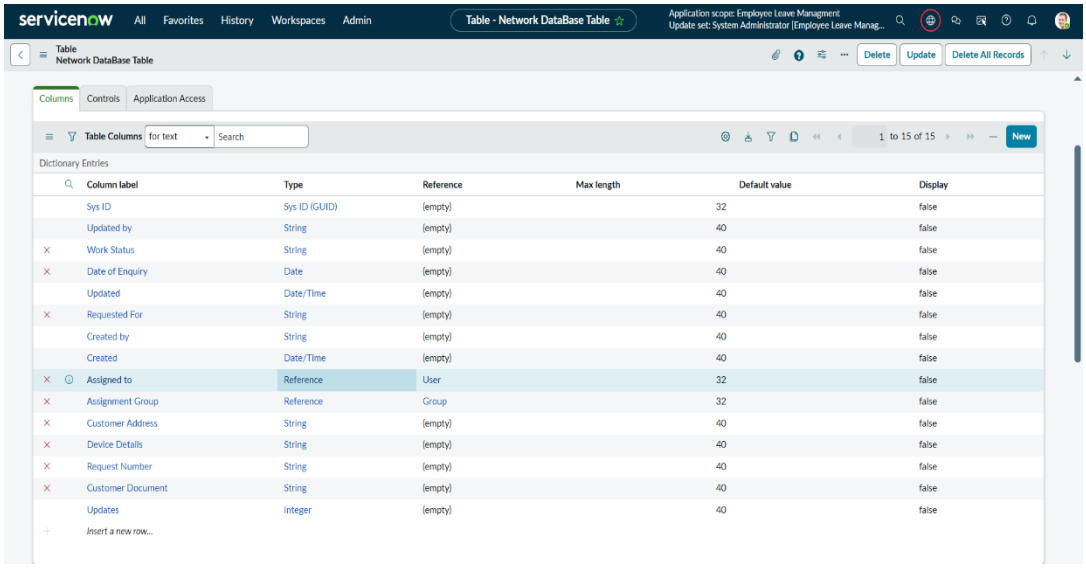
After selecting the table, you'll be brought to a view that lists all the columns (fields) that currently exist on that table.

To create a new field (column), go to the **Columns** tab (this is where all fields for the selected table are listed).

c. Create a New Field

1. In the Columns tab, click the New button located at the top-right corner of the page to create a new field.

2. You'll now be prompted with a form where you need to define the new field. The following fields need to be filled out:



c. Request Approvals Creation(Related List)

i. Creation of Related List

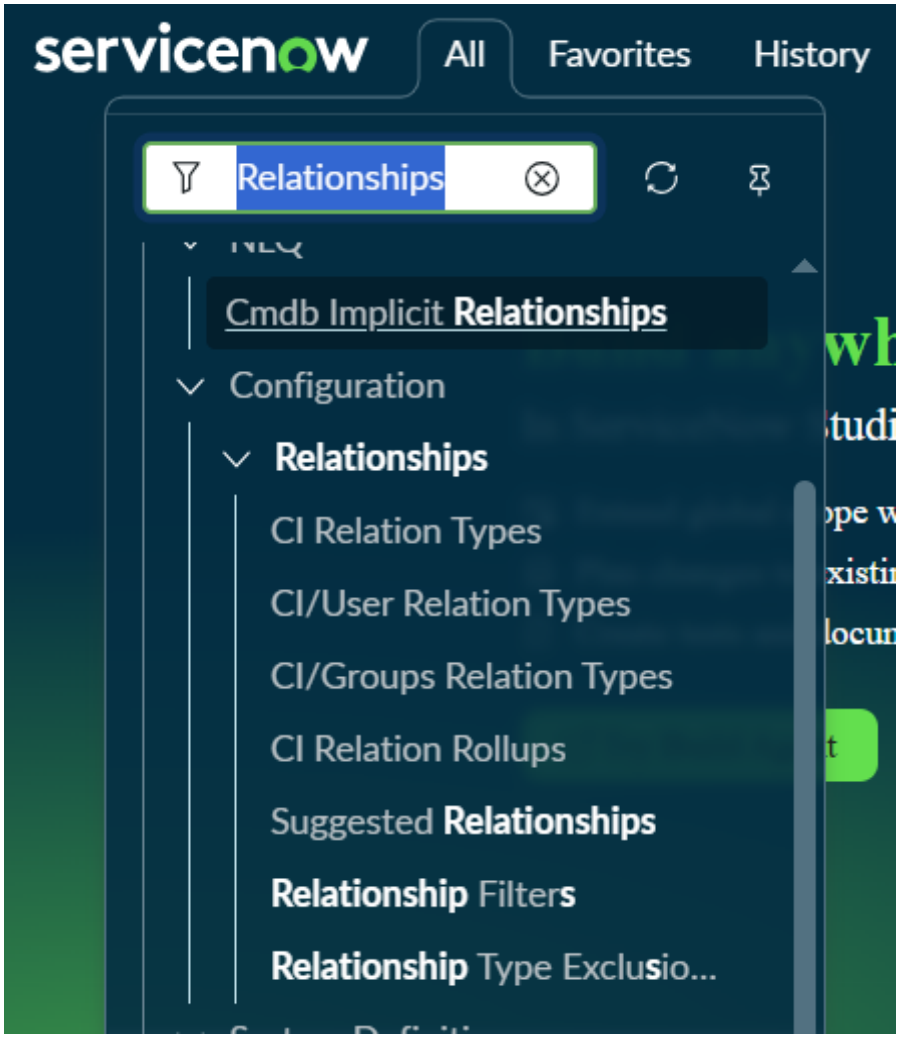
1.Navigate to **System Definition** > **Relationships.**

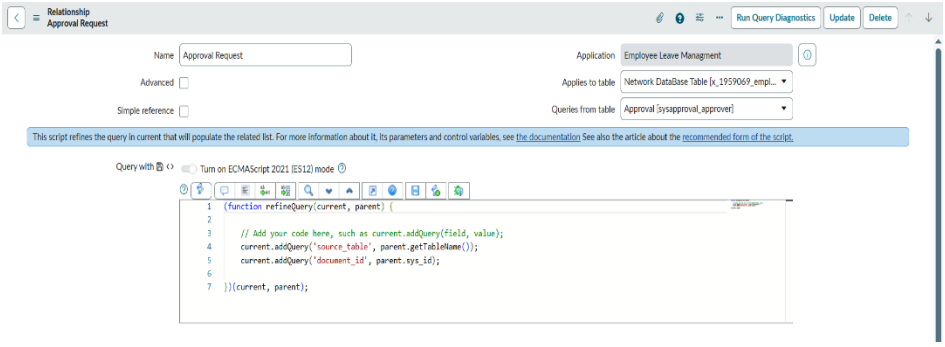
2.Click **New** to create a new relationship.

3.Fill in the following details:

- **Name**: Approval Request
- **Applies to Table** : Network Database table.
- **Queries from Table** : Sysapprovals table.
- **Active**: Make sure it's set to True.

3. Save the relationship.





ii. Adding Related List to the Table

1. Navigate to **Form Designer** for the table where you want to show related records.
2. Add a **Related List** widget to the form.
3. Select the **Related List** you want to show.

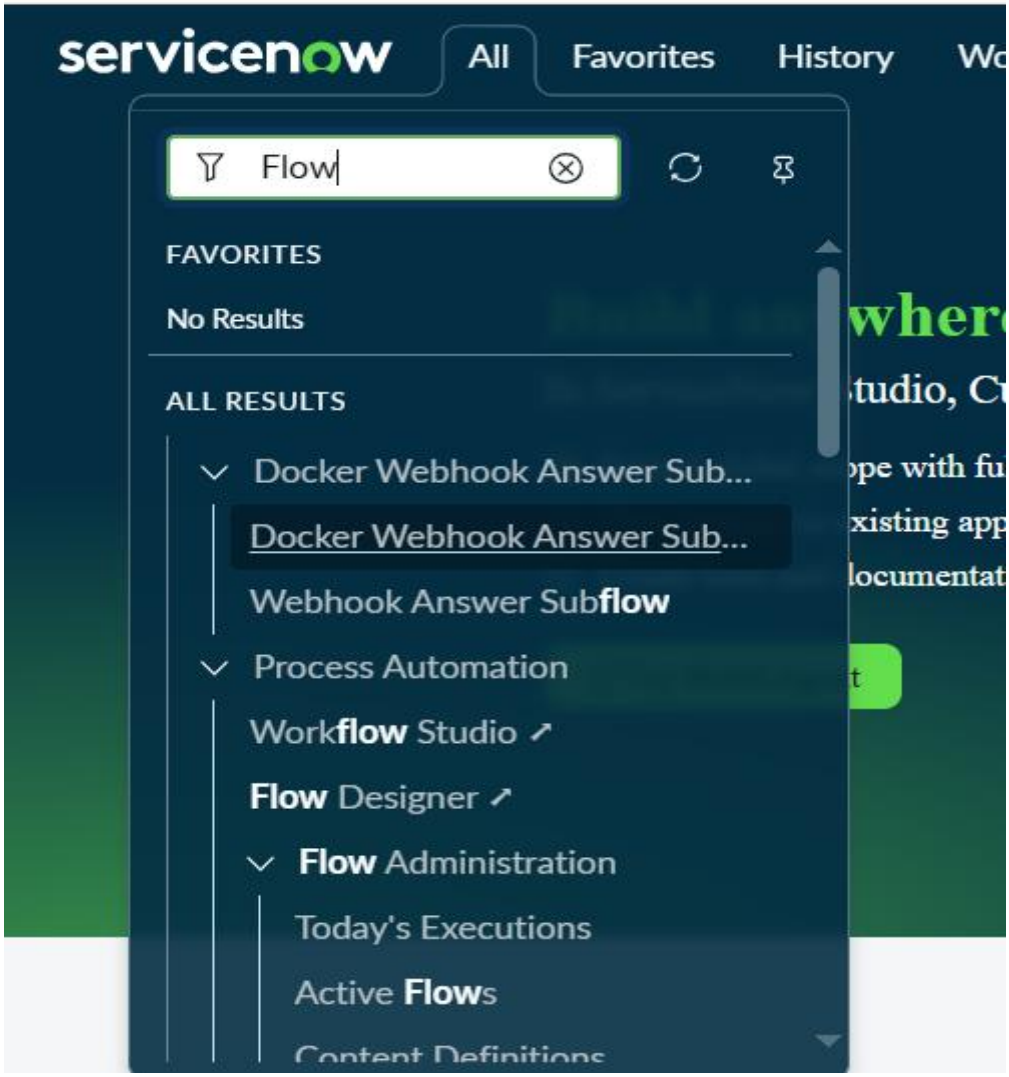
d. Creation & Implementation of flows, Actions in Flow Designer

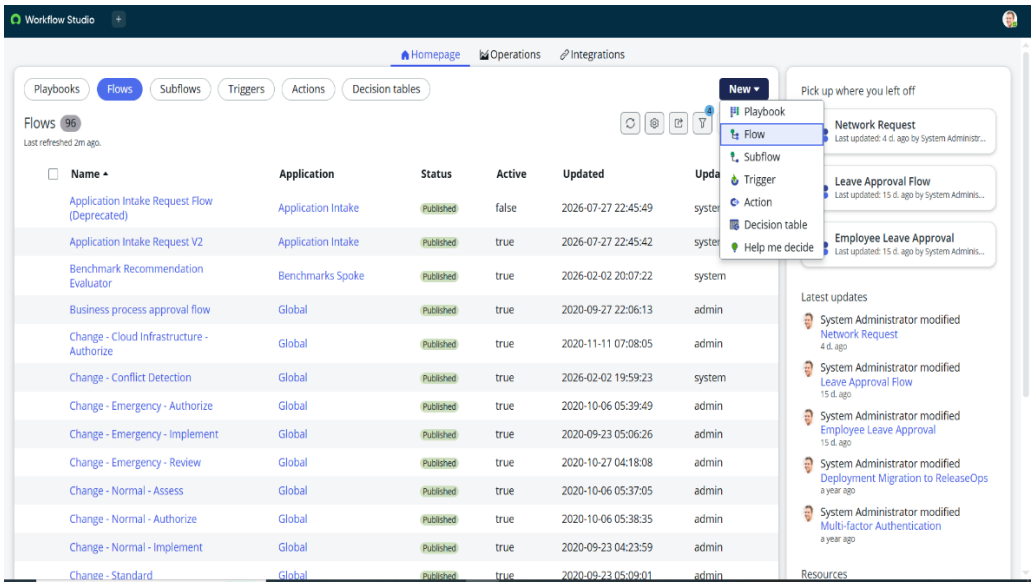
i. Creation of Flow

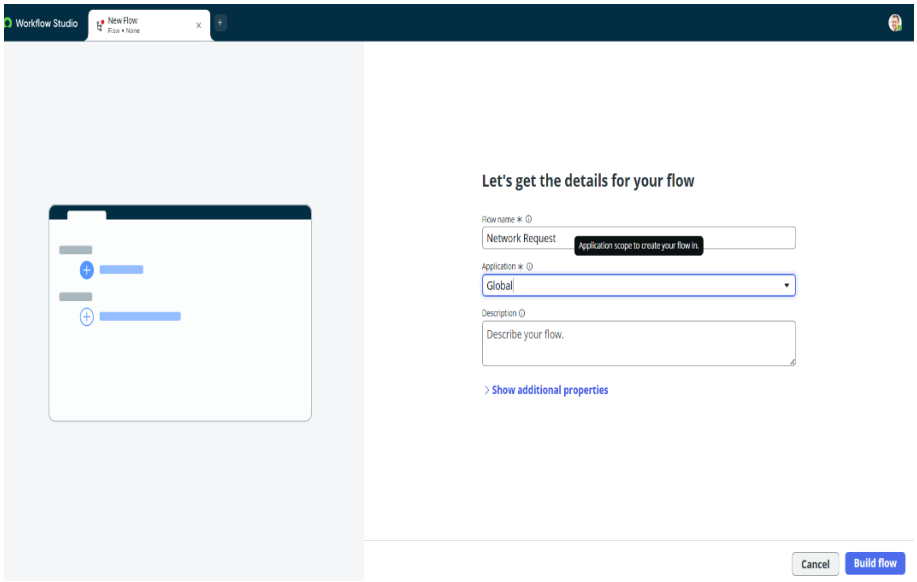
1.Navigate to Flow designer home page
2.Click on New to create a new flow
3.Provide flow name as **Network Request**
4.Provide description of flow
5.Click on Build flow.

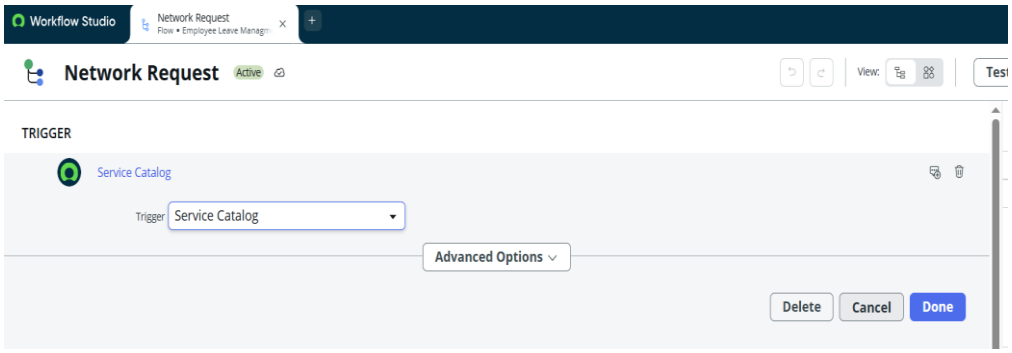
ii. Configuring Trigger

1.Click on (+) Icon to Configure the Trigger
2.Select Trigger as Application >> Service catalog
3.Click on **Done.**


servicenow
All
Favorites
History
Flow
FAVORITES
No Results
ALL RESULTS
Docker Webhook Answer Sub...
Docker Webhook Answer Sub...
Webhook Answer Subflow
Process Automation
Workflow Studio
Flow Designer
Flow Administration
Today's Executions
Active Flows


Workflow Studio
Homepage
Operations
Integrations
Playbooks
Flows
Subflows
Triggers
Actions
Decision tables
New
Playbook
Flow
Subflow
Trigger
Action
Decision table
Help me decide
Flows 96
Last refreshed 2m ago.
Name
Application
Status
Active
Updated
Application Intake Request Flow (Deprecated)
Application Intake
Application Intake Request V2
Benchmark Recommendation Evaluator
Benchmarks Spoke
Business process approval flow
Global
Change - Cloud Infrastructure - Authorize
Change - Conflict Detection
Change - Emergency - Authorize
Change - Emergency - Implement
Change - Emergency - Review
Change - Normal - Assess
Change - Normal - Authorize
Change - Normal - Implement
Change - Standard
Pick up where you left off
Network Request
Leave Approval Flow
Employee Leave Approval
Latest updates
System Administrator modified Network Request
System Administrator modified Leave Approval Flow
System Administrator modified Employee Leave Approval
System Administrator modified Deployment Migration to ReleaseOps
System Administrator modified Multi-factor Authentication
Resources



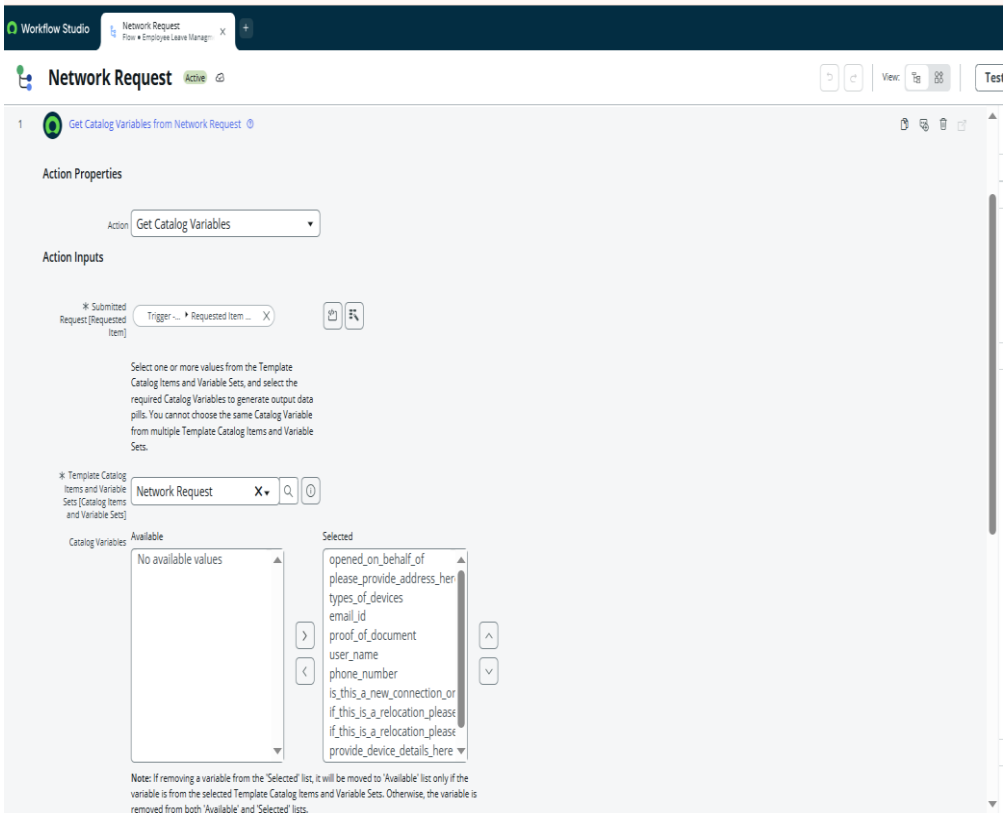


iii. Configuring Actions

Click on Actions button to configure new action

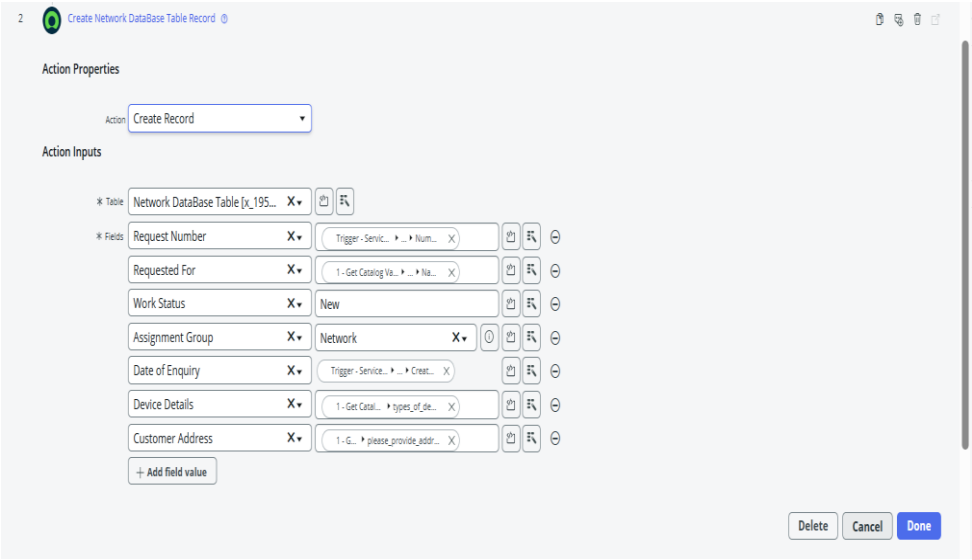
1. Get Catalog Variables

- Click on Action, search for Get Catalog Variables
- Select Get Catalog Variables
- Action Inputs>> Trigger>>service catalog>>Requested Item
- Template catalog items >> Select table >> Network Request
- Select the Required Variables and Move to the selected area.
- Click on done



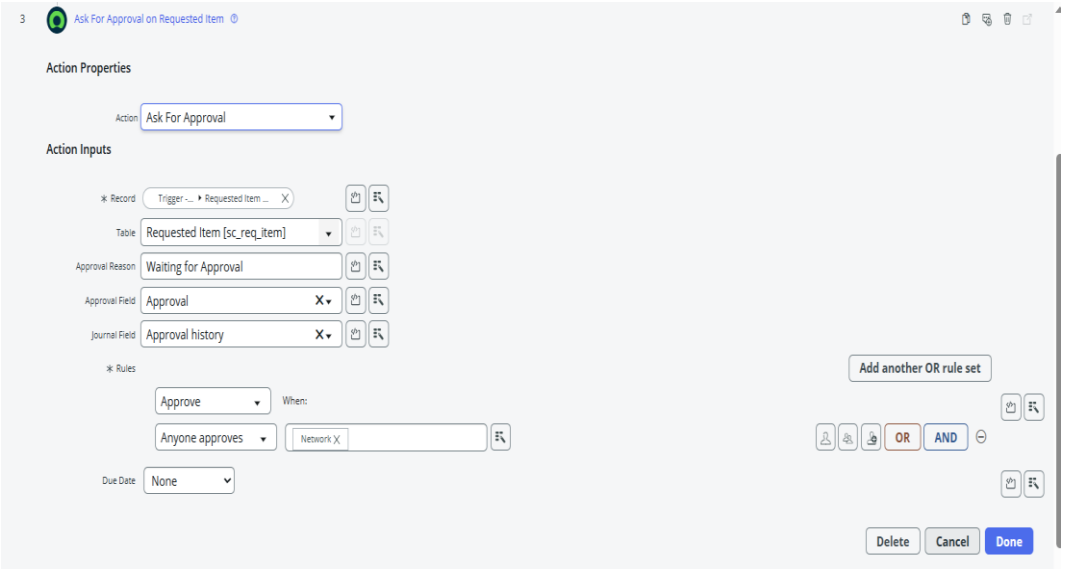
2.Create Record

- Select action as Create Record
- Select table as Network Database
- Click on Add fields button to configure the fields
- Configure the Required fields as shown in the below picture
- Click on done



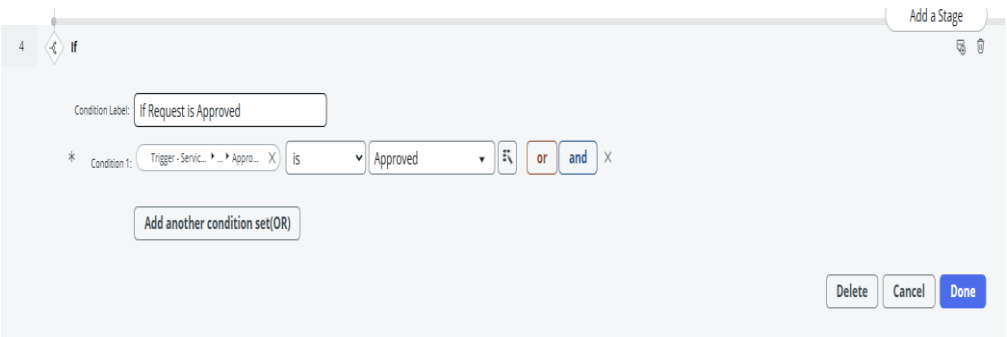
3. Ask for approvals

- Select action as Ask for Approval
- Select target record >> Create record>> network database table
- Provide Approval Reason>> Waiting for approval
- Configure approval rules>> Approve, reject, approve/reject
- Select approvals as Anyone approves, everyone approves etc.
- We can select approvals like static/dynamic as shown below
- Click on done



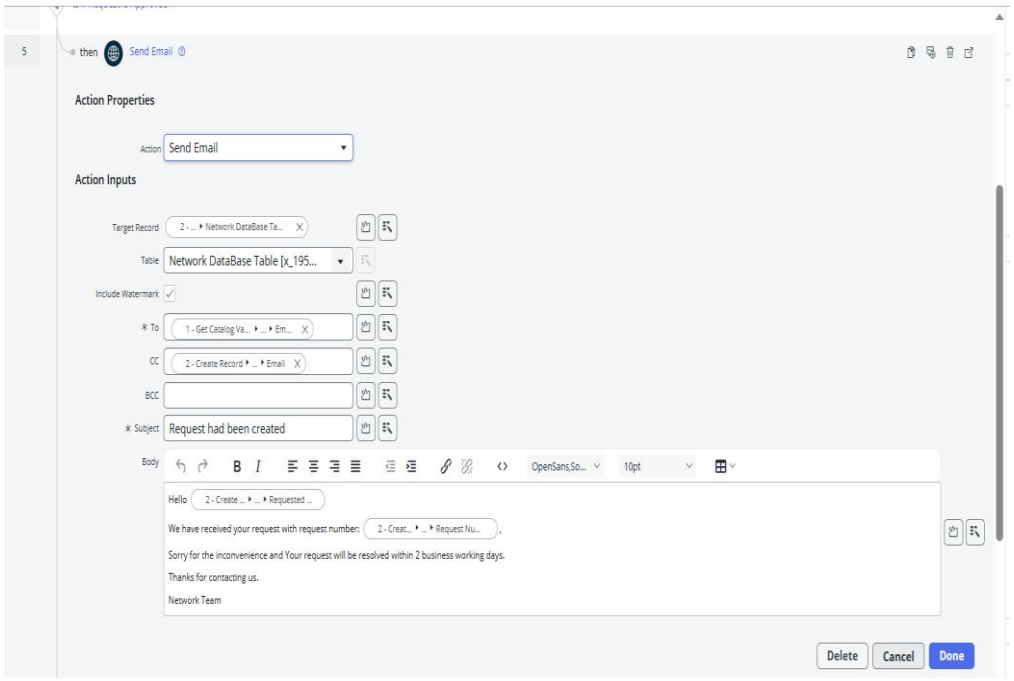
4. Flow Logic

- Select action as flow logic and Select If condition
- Apply condition >> Ask for approvals state is Approved/Rejected as per requirement
- Click on done



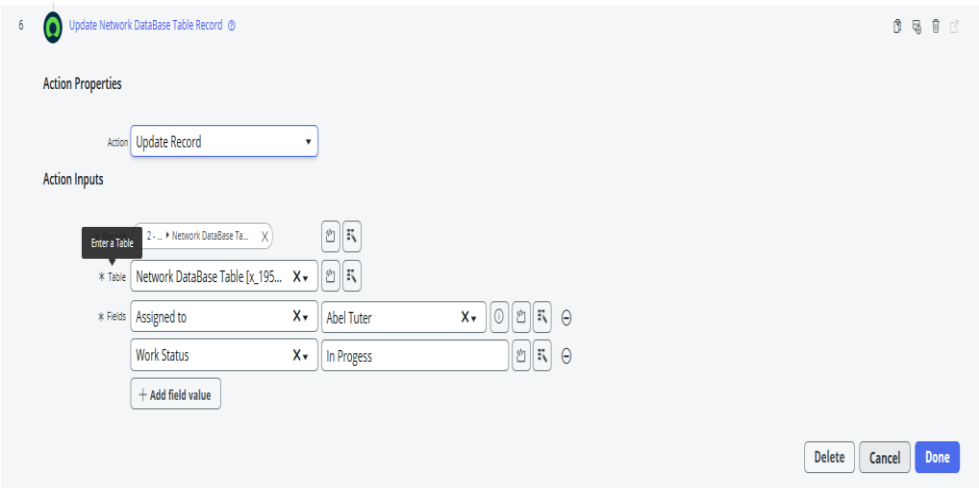
5. Send Email

●Select action as Send Email
●Select target record >> Create record>> network database table
●Table will be selected automatically
●Cofigure To, CC, BCC as per our requirements(select static/dynamic)
●Provide Subject & Body as shown in the below picture
●Click on done

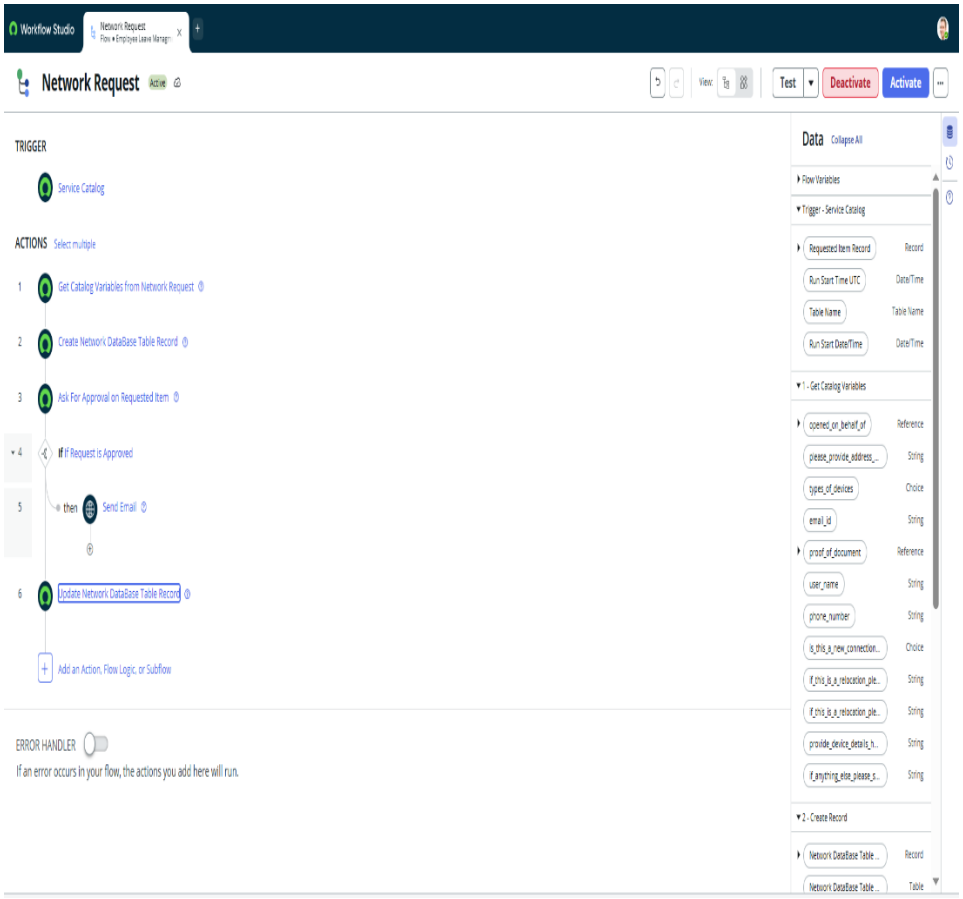


6.Update Record

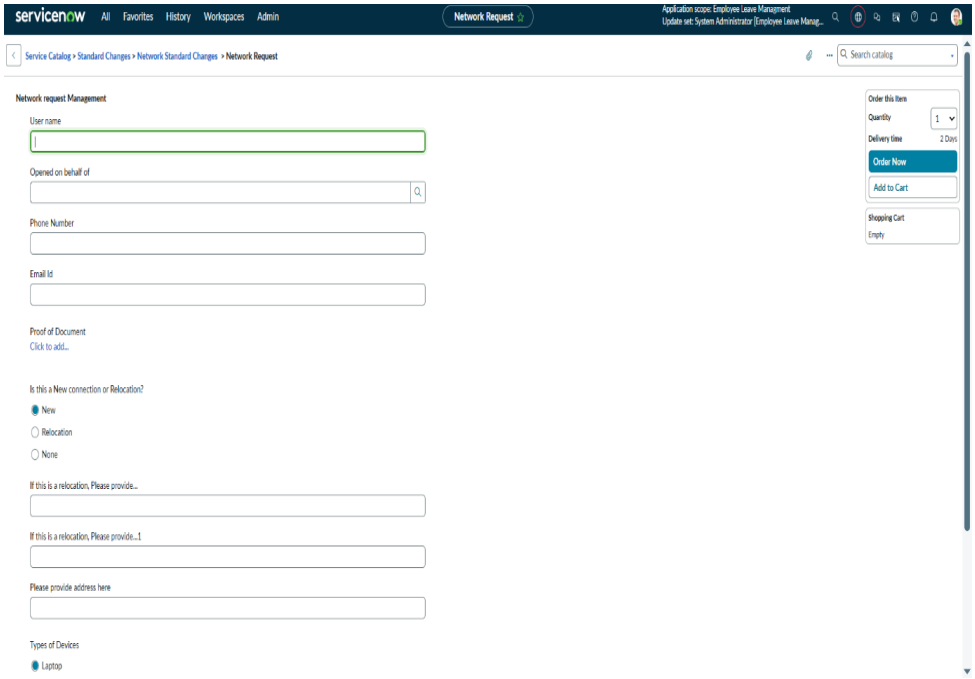
●Select action as Update Record
●Select record as >> create record>> network database
●Table will be selected automatically
●Configure the fields as per requirement, as shown in below
●Click on done



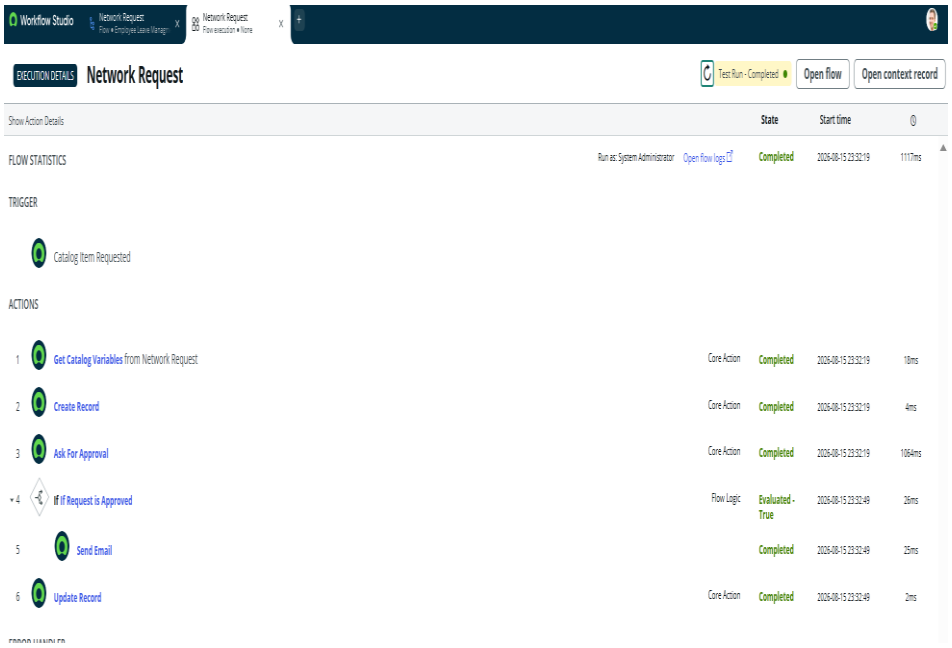
iv. Flow Chart



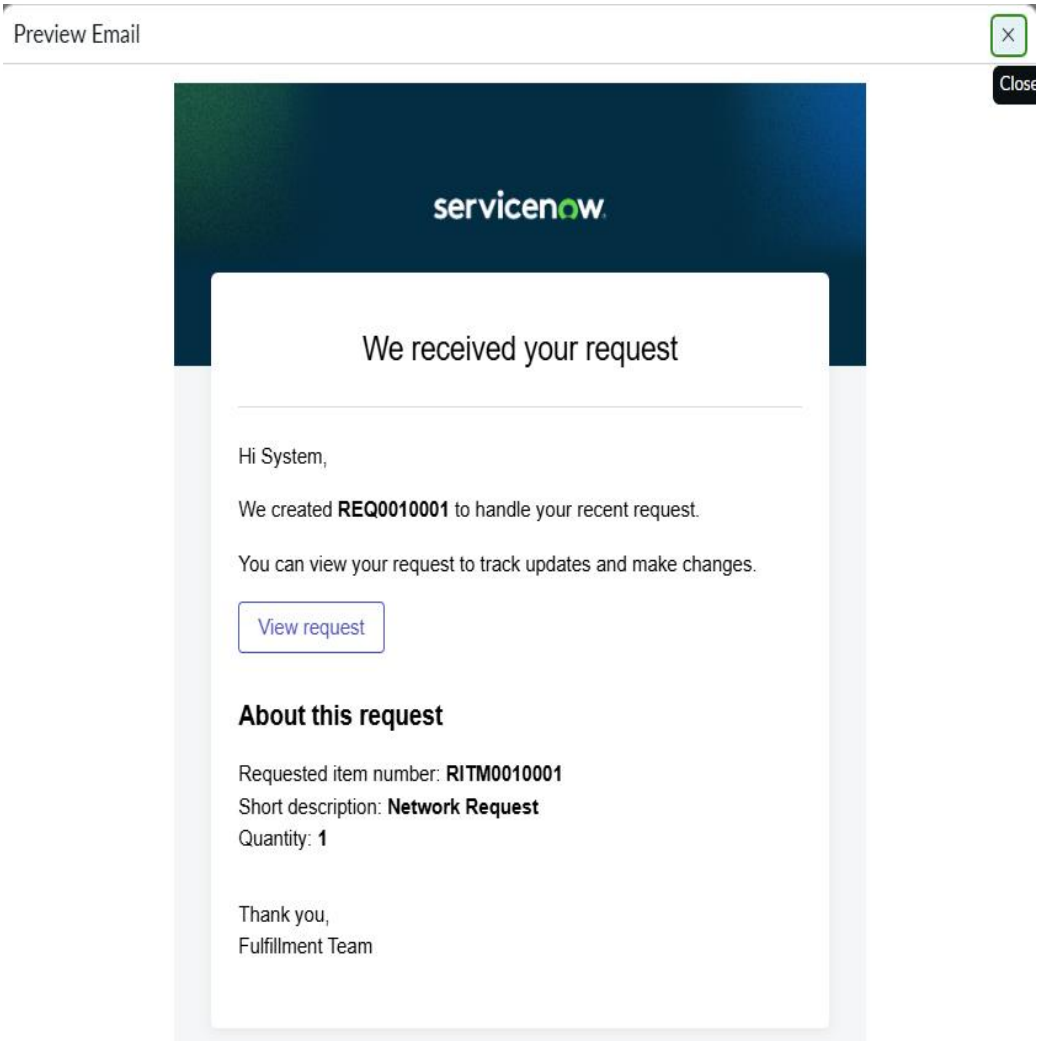
V. Screenshots of Output



VI. Execution Details



VII. EMAIL



VIII. Conclusion

The Automated Network Request Management in ServiceNow project provides a streamlined, reliable, and user-friendly solution for handling network-related service requests. By automating approvals, routing, and notifications, the system eliminates delays and errors associated with manual processes while offering transparency through real-time updates and reporting. This initiative not only improves SLA compliance and user satisfaction but also optimizes IT operations by reducing repetitive workload and enabling faster fulfillment. Ultimately, it supports the organization's goal of delivering efficient, high-quality, and responsive network services, empowering both end users and IT teams.